Understanding Social Structure in Baboon Proximity Networks

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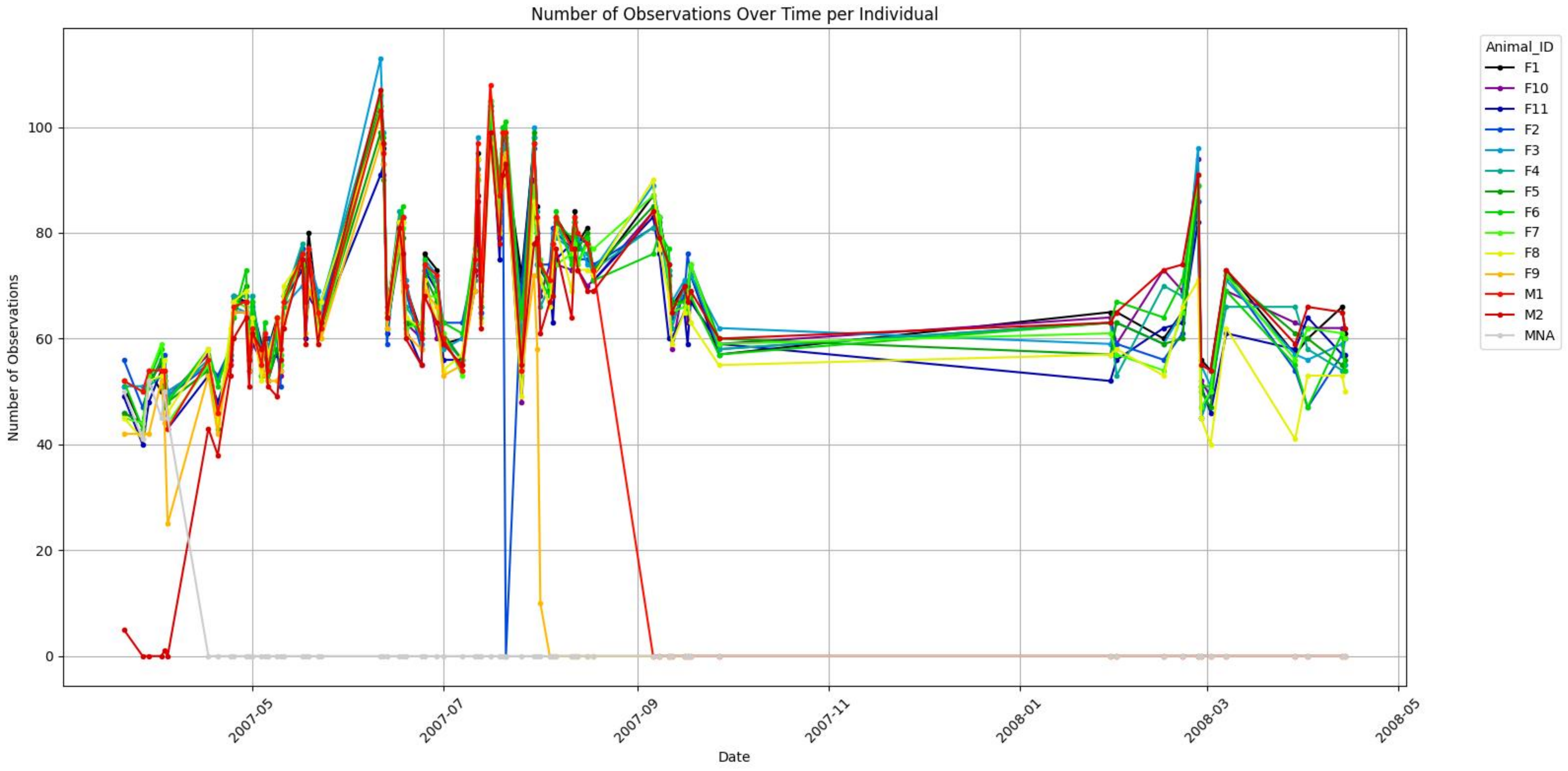
Data

- Data by Bonnell et al.
- 13 baboons, 61841 observations
- Mar. 2007 – Apr. 2008
- Attributes: date, time, coordinates, animal ID
- Animal ID: sex + dominance rank (e.g., M1 is the most dominant male)
- Dominance ranks are assigned based on the outcomes of agonistic contests among individuals

Research focus

- Correlation between metrics in proximity network and dominance hierarchy of baboons
- Persistance of social signatures (distribution of time spent with connections and identity of connections) over time (Saramäki et al.) and its relationship with dominance hierarchy in baboons

Data



Number of Observations Over Time per Individual
Number of Observations
0
20
40
60
80
100
Date
2007-05
2007-07
2007-09
2007-11
2008-01
2008-03
2008-05
Animal_ID
F1
F10
F11
F2
F3
F4
F5
F6
F7
F8
F9
M1
M2
MNA

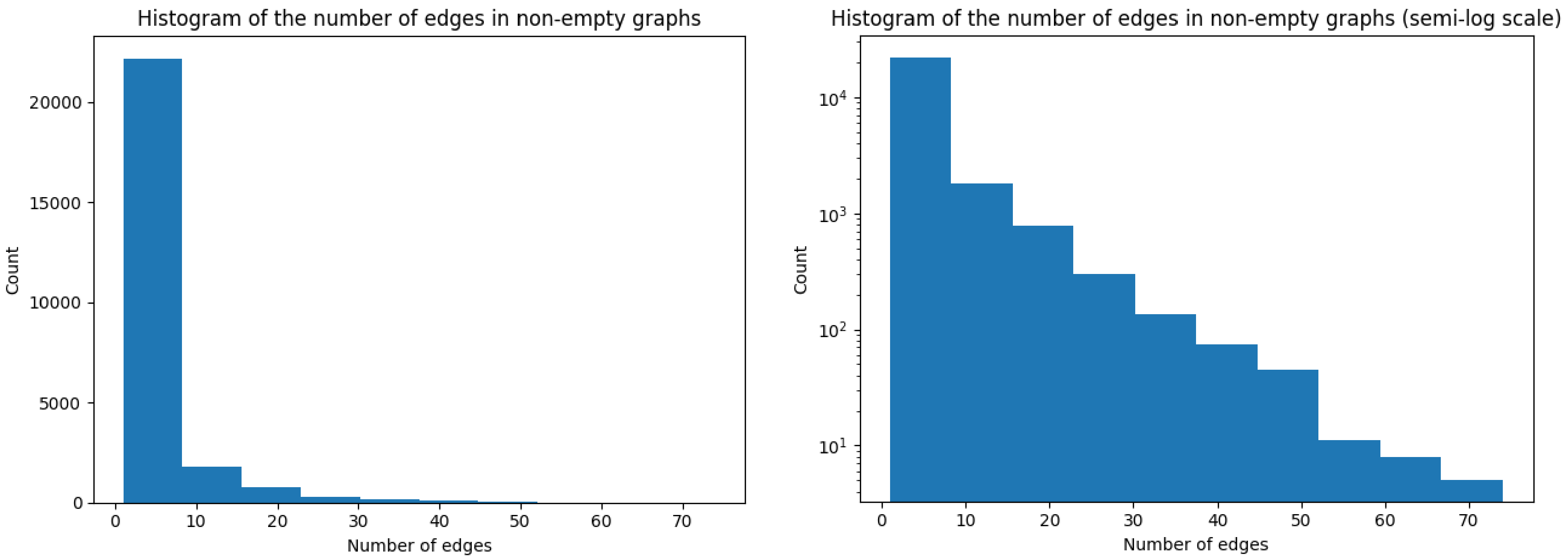
Data interpolation

- For each available time point in the dataset:
  - If the time interval between the two closest observations is at most 10 mins -> linear interpolation
  - For $t_0 < t < t_1$ and $t_1 - t_0 \leq 10$ mins:

$$\mathbf{x}_t = \frac{\mathbf{x}_1 - \mathbf{x}_0}{t_1 - t_0}(t - t_0) + \mathbf{x}_0$$

Proximity network generation

- For each time point in the dataset, if the distance between 2 baboons are at most 5m, add an edge between the 2 corresponding nodes
  -> 40539 graphs



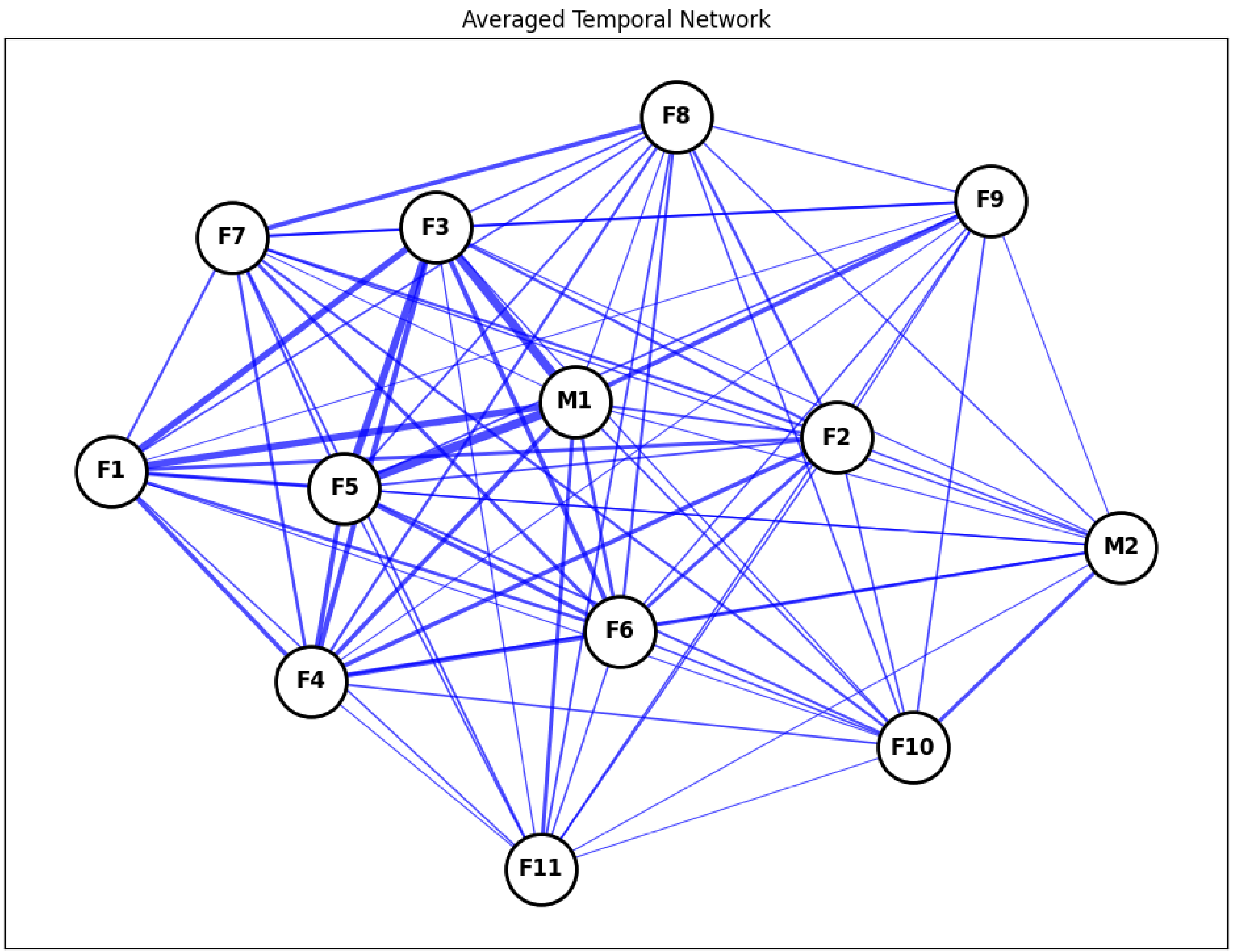
Average network

- Edge weight:

$$w_{ij} = \frac{\sum_{t=1}^{N_{ij}} \mathbb{1}[(i,j) \in E_t]}{N_{ij}}$$

where E_t is the set of edges in the graph at time point t and N_{ij} is the number of time points where there are (interpolated) observations for both i and j

Average network



Correlation of node metrics with rank

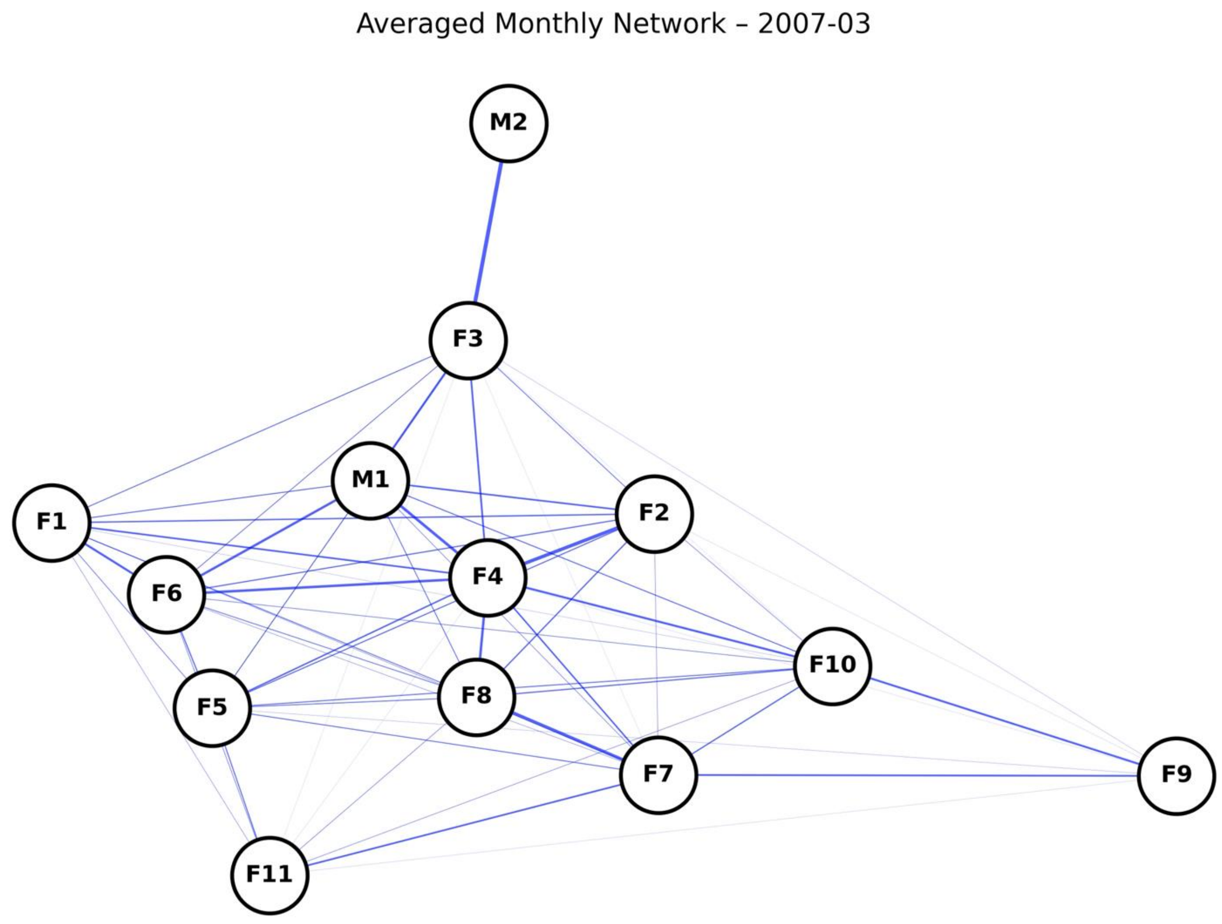
- Spearman's rho, P-values from permutation tests (1000 permutations)
- Female nodes only (since there are only 2 male nodes, and male and female rankings are not comparable)
- Higher-ranked female nodes have higher strength, clustering, closeness, and eigenvector central

Metric	**Spearman's rho**	**P-value**
Strength	-0.763636	**0.003**
Clustering coefficient	-0.763636	**0.003**
Betweenness centrality	-0.074536	0.466
Closeness centrality	-0.727273	**0.002**
Eigenvector centrality	-0.818182	**0.000**

Correlation of edge weight with rank difference

- Spearman's rho between edge weights and the absolute differences between the ranks of endpoint nodes = -0.559, P = 0.000
  -> Baboons with larger differences in social ranks tend to spend less time together

Average network by month



Average network by temporal block

- 3 blocks:
  - Block 1: Mar 2007 – Jun 2007
  - Block 2: Jul 2007 – Sep 2007
  - Block 3: Jan 2008 – Apr 2008

Ego plots

- Ego network of node *i*: subgraph with only edges adjacent to *i*
- Fraction of strength of node *j* in the ego network of node *i*:

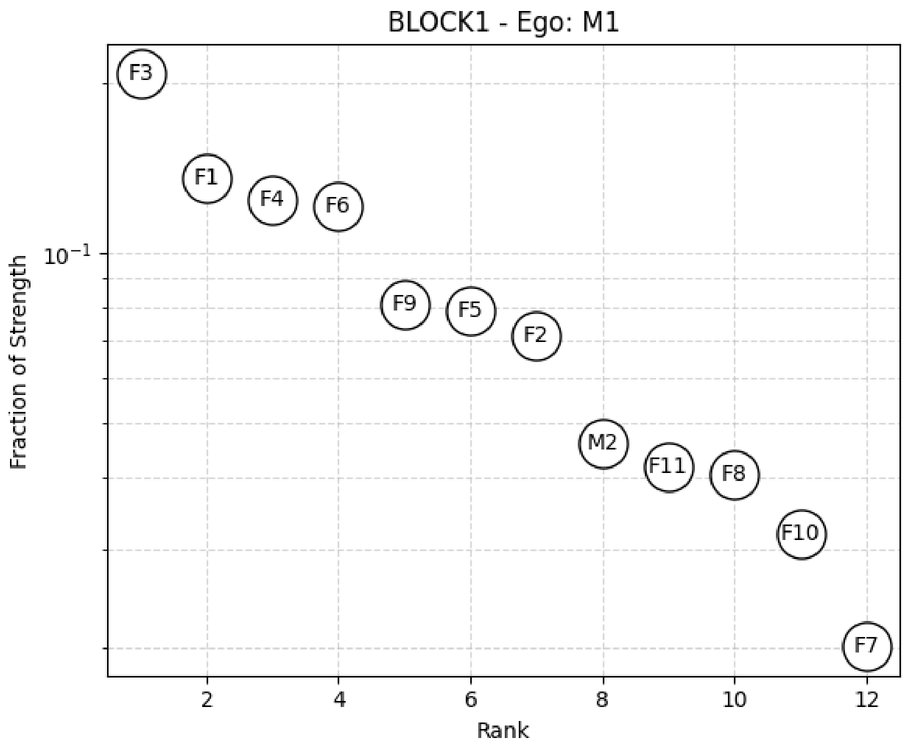
$$\frac{w_{ij}}{\sum_{j' \in N(i)} w_{ij'}}$$

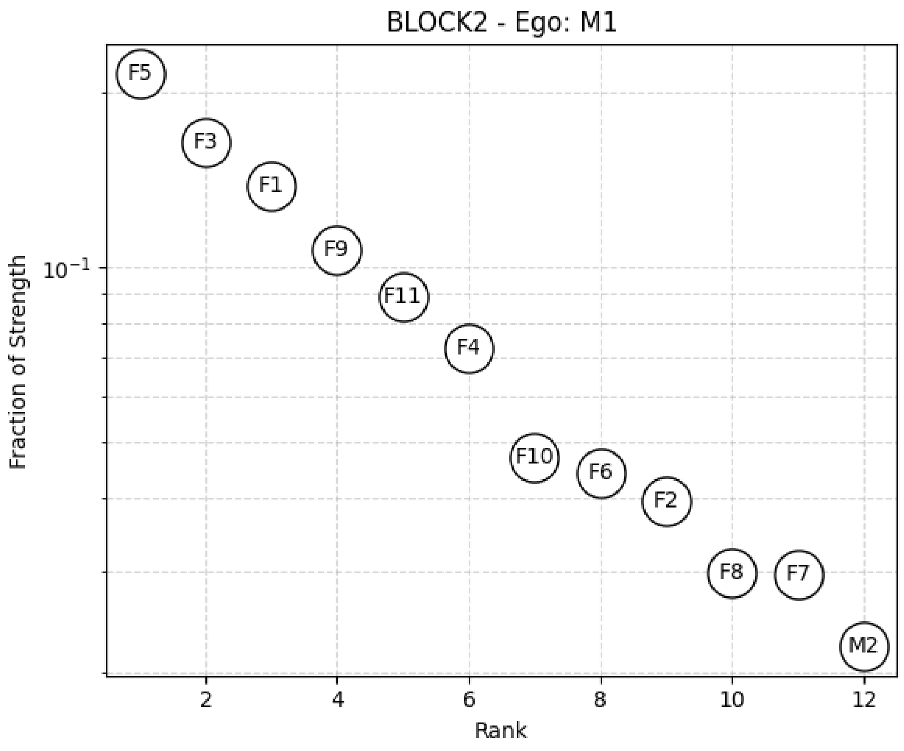
where $N(i)$ is the set of neighbors of node i

- Represents the proportion of time node *i* spends with node *j* relative to the time *i* spends with all other nodes

Ego plots: M1

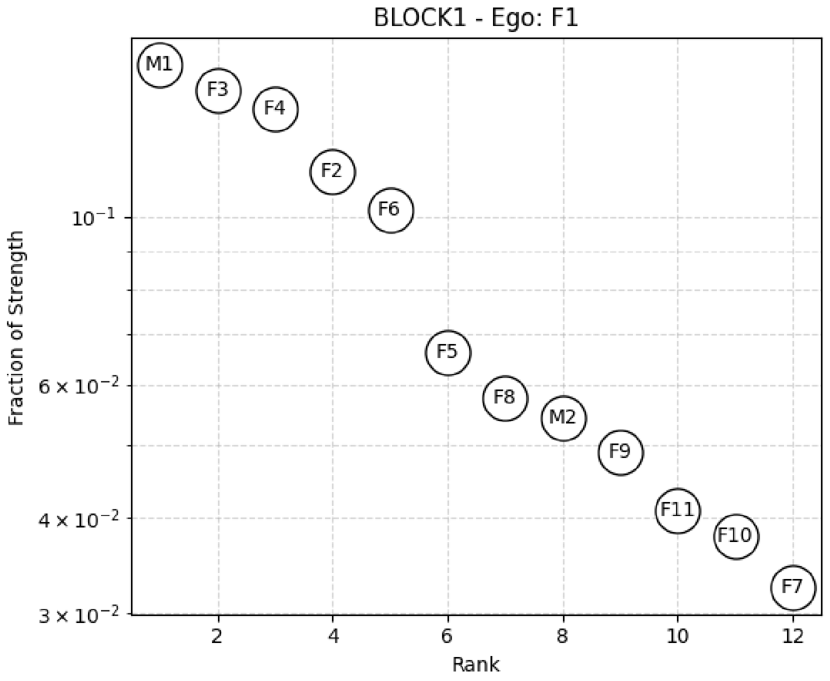
- No observation of M1 in block 3

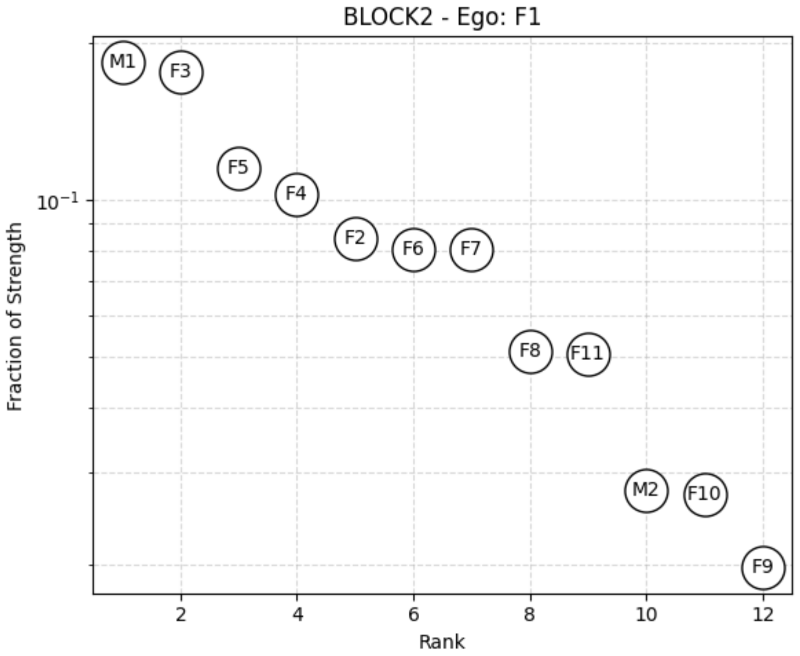


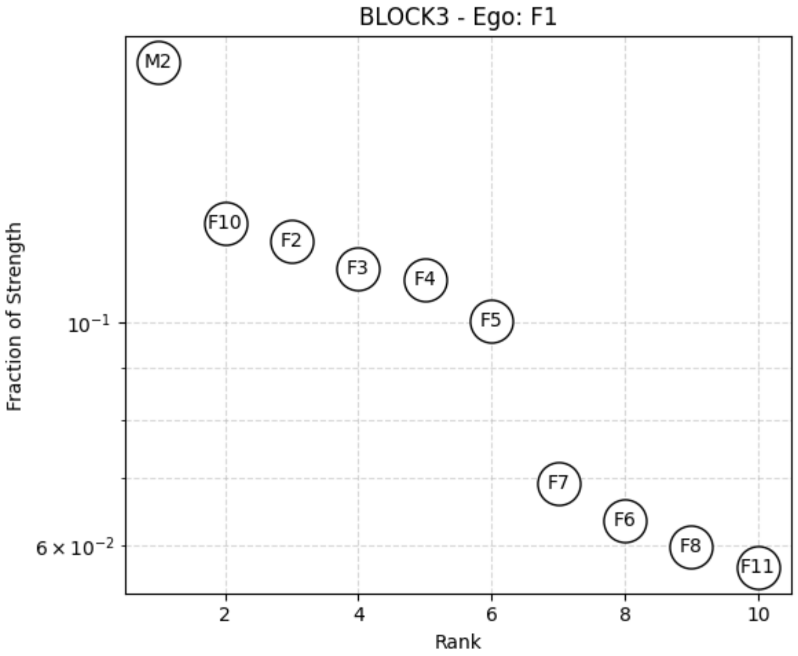


Ego plots: F1

- F1 spends the most time with M1 in block 1 and 2 and with M2 in block 3

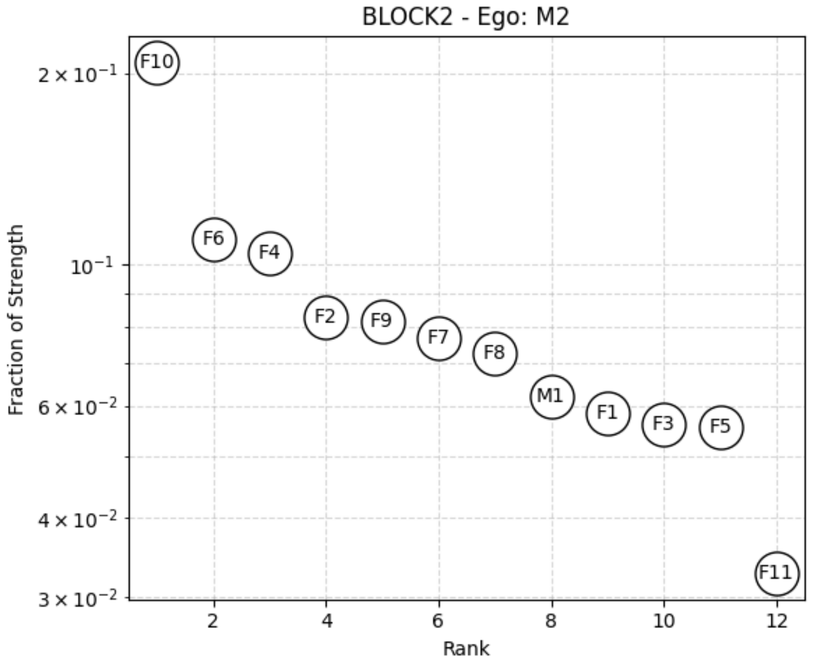


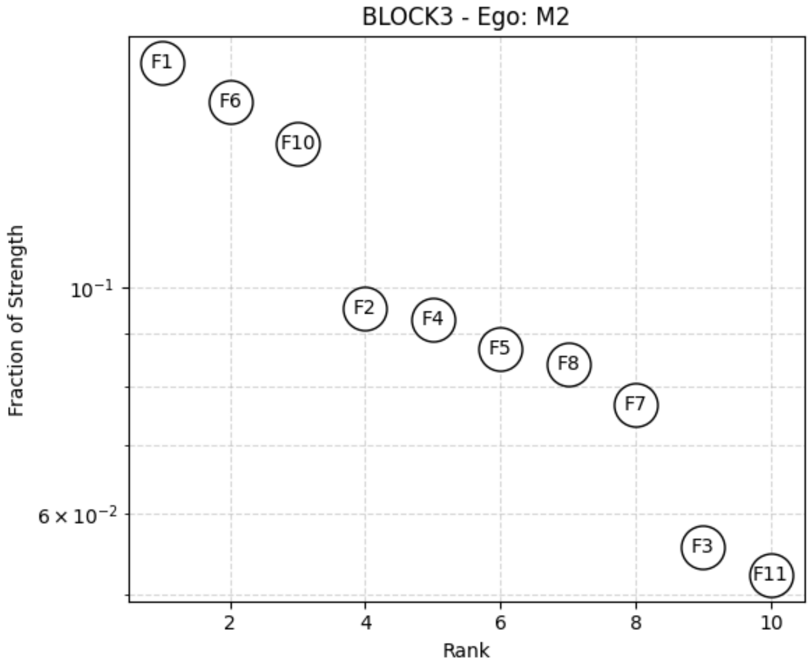




Ego plots: M2

- M2 spends the most time with lower-ranking females (F6, F10, etc.) in block 1 and 2 and with F1 in block 3





Persistance of top neighbors

- Identify top 5 neighbors of each node in each block
- Notation:

$J_{b_n,b_k}(i,j)$ is the Jaccard index between top neighbors of node i in block b_n and node j in block b_k

- Calculate Jaccard index of top neighbors of each node in consecutive blocks (block 1 – block 2 and block 2 – block 3)
- Average over nodes:

$$\frac{1}{N}\sum_{i=1}^{N} J_{b_1 b_2}(i,i) = 0.493 \pm 0.235$$

$$\frac{1}{N}\sum_{i=1}^{N} J_{b_2 b_3}(i,i) = 0.391 \pm 0.160$$

Correlation of top neighbors retention with rank

- For each node, calculate the top neighbor retention

$$\frac{1}{2}J_{b_1 b_2}(i,i) + \frac{1}{2}J_{b_2 b_3}(i,i)$$

- Correlation with rank (females only): Pearson's rho -0.844, P = 0.001 -> higher-ranked female baboons tend to retain social connections better

Persistance of top neighbors

- Compare the average self-distance and baseline (non-self):

$$\text{mean}_{\,i\in\{1,\dots,N\}}\left(\frac{1}{2}J_{b_1b_2}(i,i)+\frac{1}{2}J_{b_2b_3}(i,i)\right)=0.445\pm0.128$$

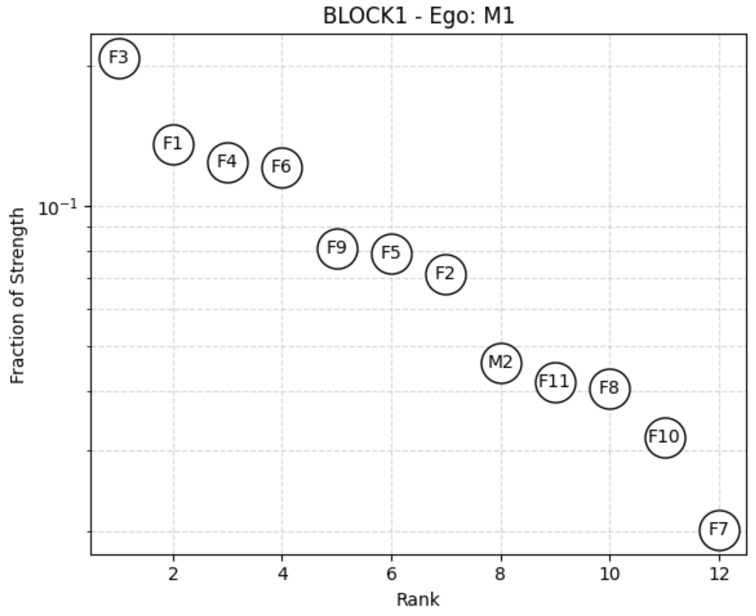
$$\text{mean}_{\,i,j\in\{1,\dots,N\},i\neq j}\left(\frac{1}{2}J_{b_1b_2}(i,j)+\frac{1}{2}J_{b_2b_3}(i,j)\right)=0.272\pm0.113$$

- P-value of difference: 0.000
  -> self-distance tend to be smaller, baboons tend to retain their top connections compared to baseline

Persistance of ego plot shape

- Shape of the ego plot: distribution of time spent with neighbors (ego distribution)
- Jensen-Shannon divergence (JSD) to measure the shape difference between 2 ego plots
- Notation:

$JSD_{b_n,b_k}(i,j)$ is the JSD between the ego distribution of node i in block b_n and node j in block b_k



Correlation of ego plot shape retention with rank

- For each node, calculate the ego plot shape retention

$$\frac{1}{2}JSD_{b_1b_2}(i,i) + \frac{1}{2}JSD_{b_2b_3}(i,i)$$

- Correlation with rank (females only): Spearman’s rho 0.200, P = 0.567 -> not significant

Persistance of ego plot shape

- Compare the average self-distance and baseline (non-self):

$$\text{mean}_{\ i\in\{1,\ldots,N\}} \left(\frac{1}{2}JSD_{b_1b_2}(i,i)+\frac{1}{2}JSD_{b_2b_3}(i,i)\right)=0.119\pm 0.028$$

$$\text{mean}_{\ i,j\in\{1,\ldots,N\},i\neq j} \left(\frac{1}{2}JSD_{b_1b_2}(i,j)+\frac{1}{2}JSD_{b_2b_3}(i,j)\right)=0.133\pm 0.022$$

- P-value of difference: 0.018
  -> self-distance tend to be smaller, baboons tend to retain their ego plot shape compared to baseline

Summary of key findings

- Higher-ranked baboons are more socially central (higher strength, clustering, closeness, and eigenvector central)
- Baboons with larger differences in social ranks tend to spend less time together
- Higher-ranked baboons tend to better retain social connections over time, but there's no correlation between ranking and the persistence of the distribution of time spent with connections
- Baboons tend to maintain social signatures over time, both in the distribution of time spent with connections and the identity of those connections

Reference

Bonnell, T. R., Clarke, P. M., Henzi, S. P., & Barrett, L. (2017). Individual-level movement bias leads to the formation of higher-order social structure in a mobile group of baboons. *Royal Society Open Science*, **4**, 170148. https://doi.org/10.1098/rsos.170148

Saramäki, J., Leicht, E. A., López, E., Roberts, S. G. B., Reed-Tsochas, F., & Dunbar, R. I. M. (2014). Persistence of social signatures in human communication. *Proceedings of the National Academy of Sciences of the United States of America*, **111**(3), 942–947. https://doi.org/10.1073/pnas.1308540110